

# Mohamed Ahmed

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## Education

### University of Minnesota

Expected May 2027

*Bachelor of Science in Electrical Engineering*

**Relevant Coursework:** Linear Control Systems & Lab, Power Electronics, Embedded Systems, Analog Electronics, Wireless Hardware Design

## Experience

### Michels Power

May 2026 – Aug 2026

*Project Management Engineering Intern*

- Supported electric utility projects through field verification, GIS records, engineering drawings, and technical documentation.
- Reviewed engineering drawings and as-built records for transmission and distribution projects.
- Developed a technical database tracking **1,200+ utility as-built records** for project documentation.

## Club Experience

### STRIVE Medical

September 2025 – Present

*Electrical / Hardware Team Lead*

- Lead electrical development for a **medical robotics platform** integrating sensing, actuation, embedded control, and power electronics.
- Integrated **Teensy 4.1, ESP32, 14-bit encoders, IMUs, and motor controllers** across robotic subsystems.
- Developed embedded firmware for **sensor acquisition and motor-control interfaces** across the robotic platform.
- Designed a **4-layer PCBA** supporting motor-drive hardware, embedded controllers, sensing, and power distribution.
- Brought up and debugged electromechanical hardware using oscilloscopes and logic analyzers; led electrical development to **unanimous 1st Place** at the 2026 UMN MVP Challenge.

## Projects

### Feedback & Motor Control | MATLAB, Simulink, Quanser

- Modeled the dynamics of a **DC motor** and developed MATLAB/Simulink models for feedback-control analysis.
- Designed and tuned **PI speed controllers** and evaluated closed-loop stability, transient response, and robustness on a Quanser DC Motor Control Trainer.
- Designed and tuned **PID position controllers** and investigated integral windup under actuator constraints.
- Performed **system identification** and developed feedback control for a nonlinear magnetic-levitation plant.

### 100 W Synchronous Buck Converter | Feedback Control, Dynamic Modeling, Embedded Control

- Developed averaged and **small-signal models** of a 100 W, 24 V-to-12 V synchronous buck converter to characterize system dynamics.
- Derived the **control-to-output transfer function** and analyzed frequency response and stability using MATLAB.
- Designed a **PI voltage controller** using Bode analysis, crossover frequency, phase margin, and transient-response objectives.
- Used **MATLAB and LTspice** to compare analytical control models with switching-circuit behavior and transient response.
- Designed an STM32G4-based digital control architecture using **PWM generation, ADC feedback, voltage/current sensing, and gate-drive circuitry**.

## Skills

**Controls:** Feedback Control, PI/PID Control, Transfer Functions, Bode Analysis, Stability Margins, System Identification

**Motor Control & Mechatronics:** DC Motor Modeling, Speed Control, Position Control, Encoders, Motor Controllers, Sensor Feedback

**Modeling & Simulation:** MATLAB, Simulink, LTspice, Dynamic Modeling, Small-Signal Modeling, Frequency-Domain Analysis

**Embedded Control:** STM32G4, ESP32, Teensy 4.1, Embedded C/C++, PWM, ADC, Interrupts

**Hardware & Test:** PCB Design, Sensor Interfaces, Oscilloscope, Logic Analyzer, Multimeter, PCB Bring-Up

**Tools:** MATLAB, Simulink, LTspice, Altium Designer, KiCad, Git, SolidWorks